IRT / RTSM User Acceptance Testing

Cheat-sheet · operations-only · synthetic · no PHI · say **participant** not **subject**

What UAT confirms

In a test environment you confirm the vendor's randomization and drug-supply build does **exactly** what your URS and specs say, before any real participant is randomized. You operate the staging IRT in a browser, capture dated screenshot evidence, and transcribe results into your workbook so the record is attributable and stands alone for an auditor.

The go-live gate (conjunctive)

Sign-off is READY only when **all three** hold:

- Every **Critical** test passes
- **0** open Critical / Major defects
- **100%** traceability (every requirement has a test)

Snapshot now: **NOT ready - blocked**.

Severity, not count, decides. The gate packages evidence; a named human signs.

Live snapshot

Test scripts	**29**
Passed	**22**
Failed	**2**
Open defects	**3**
Requirements traced	**97%**

Defect severity drives the gate

Critical	Could break the blind, mis-randomize, mis-dose, or corrupt integrity	Blocks
Major	Significant functional error in a key workflow (resupply, wrong kit, missing alert)	Blocks til fixed
Minor	Cosmetic / non-functional (label typo, missing site number)	Fix or waive

Where it runs: the systems (the workbook is the paper trail)

Staging IRT (vendor, test list) → **Operate in a browser** → **Dated screenshots + exports** → **Excel workbook** (status·defects·traceability·gate) → **eTMF** (sign the gate package)

Failures also go to the **vendor defect tracker**; the vendor promotes a fixed build back to staging for retest. The workbook never connects to the IRT, you transcribe, on purpose.

Anatomy of an executable test (RAND-02)

A good test has a **specific, verifiable** expected result, not "works correctly".

- **Precondition:** test list loaded; the <70 kg stratum, block 1, has an unused slot.
- **Steps:** register 101-001 at 68.0 kg → randomize → record kit + confirmation → open the allocation report as the unblinded statistician.
- **Expected:** allocated to the next unused <70 kg / block-1 entry; arm, sequence, kit match the list; the investigator sees a **kit ID only**, not the arm.
- **Verify:** compare cell-for-cell across every stratum and both arms, not one happy-path participant.

The 8-step workflow

1. Derive test cases from URS + specs (positive · negative · boundary)
2. Load the approved test list into staging IRT
3. Execute each case in the browser; be systematic
4. Record result + dated screenshot evidence
5. Log every failure as a defect with a severity
6. Vendor fixes; retest the new build
7. Traceability: every requirement maps to a passing test
8. Evaluate the gate; a named human signs it into the eTMF

Run the interactive tool

```
install.packages c 'shiny' 'bslib' 'DT' 'ggplot2'))
shiny::runApp 'uat_irt/shiny'
```

Select a test, mark it Pass/Fail, close a defect, and the KPIs

```
+ go-live gate re-compute live. The gate math lives in shiny/R/logic.R, shared by the app, the guide, the slides, and the unit tests. Full source in the run-and-build page.
```